📊 PROJECT DOCUMENTATION

Indian Shares

A full-stack real-time stock market research platform for NSE & BSE, built with modern web technologies.

PROJECT NAME	FRAMEWORK	DATA SOURCE
Shares App	Next.js 14 (App Router)	Yahoo Finance API

MARKET FOCUS	LANGUAGE	DATE
NSE & BSE (India)	TypeScript	March 2026

Next.js 14 · React 18 · TypeScript · Tailwind CSS · Yahoo Finance 2 · Recharts · SWR · SSE (Live Data)

SECTION 01

Project Overview

The **Indian Shares Intelligence App** is a premium, real-time stock market research platform built for Indian investors. It aggregates live market data from *NSE (National Stock Exchange)* and *BSE (Bombay Stock Exchange)* via the Yahoo Finance API, processes it through Next.js serverless API routes, and presents it through a polished, dark-themed React frontend. Users can track live indices, discover top gainers/losers, deep-dive into individual stocks, and maintain a personal watchlist.

Live Market Feed

Server-Sent Events (SSE) stream index prices (Nifty 50, Sensex, etc.) in real-time with flash animations.

Top Movers

Auto-refreshing tabs showing today's top gainers, losers, and most-trending NSE/BSE stocks.

Stock Detail Page

Full analysis view with historical price charts (Recharts), fundamentals, market cap classification, and news.

Personal Watchlist

Save and track your favorite stocks. Stored in browser localStorage for persistence without a database.

Dark / Light Theme

Full theme toggle using next-themes with CSS custom properties for smooth, flicker-free switching.

Stock Search

Instant search across 200+ Indian stocks with quick-search pills for popular symbols like RELIANCE, TCS, HDFC.

SECTION 02

Technologies Used

HTML5

Semantic Markup

JSX (HTML-in-JavaScript) is used throughout all React/Next.js components. Semantic elements like <section>, <nav>, <main>, <header> ensure accessibility and SEO.

CSS3 + Tailwind

v3.4

Tailwind CSS utility classes for rapid layout and styling. Custom CSS variables (--bg-primary, --text-muted etc.) in globals.css manage the dark/light theme system. PostCSS processes the styles.

TypeScript

v5

All source files use TypeScript (.tsx / .ts). Strict type definitions ensure correctness for stock data interfaces (StockQuote, IndexData, etc.) and API responses.

Next.js

v14.2 – App Router

Core framework providing file-based routing (App Router), Server Components, serverless API Routes, and SSR/SSE capabilities. The /api/* routes run as Node.js Edge/Serverless functions.

React

v18

Component-based UI rendering. Uses React Hooks extensively: useState for local state, useEffect for SSE lifecycle management, useCallback for memoization, and useRef for DOM refs and timers.

yahoo-finance2

v3.13

Node.js library that scrapes Yahoo Finance. Used server-side in API routes to fetch real-time quotes, historical OHLC data, market summaries, market movers, and search results for NSE/BSE stocks.

Framer Motion

v11.3

Powers all UI animations: page enter transitions, stock card hover effects (y-translation, scale), staggered list animations

Recharts

v2.12

SVG-based charting library for React. Renders the historical price chart (Area/Line chart) on the individual stock

with delay, and smooth tab transitions throughout the app.

detail page including price, volume, and time-range selectors (1W, 1M, 3M, 1Y).

SWR

v2.2

React data-fetching library by Vercel. Provides intelligent caching, revalidation on focus, and deduplication for stock data fetching in the useLivePrice custom hook.

Lucide React

v0.427

Icon library with 800+ consistent SVG icons. Used for TrendingUp, TrendingDown, Star (watchlist), Activity, RefreshCw, ArrowUpRight/Down, Zap, ChevronRight throughout the UI.

next-themes

v0.3

Handles dark/light mode switching without flash. Integrates with Next.js to persist the user's theme preference in localStorage and applies the correct CSS class on the HTML element.

Axios

v1.7

Promise-based HTTP client used as an alternative to native fetch for some API calls. Supports request interception, timeouts, and automatic JSON transformation.

clsx + tailwind-merge

Utility

Helper utilities to conditionally join CSS class names and intelligently merge conflicting Tailwind classes. Used in the cn() utility function in lib/utils.ts.

Server-Sent Events (SSE)

Web Standard

Native browser API used to stream live index price updates from the server (/api/live endpoint). Provides a persistent, one-directional connection for real-time data without WebSocket overhead.

SECTION 03

Architecture Overview

CLIENT

Browser (React 18 + Next.js App Router)

Home Page (SSE consumer) | Stock Detail Page | Top Stocks Page

Watchlist Page | Navbar + ThemeProvider

↕ HTTPS / SSE

SERVER

Next.js Serverless API Routes (/api/*)

/api/market | /api/movers | /api/live (SSE stream) | /api/search

/api/stock/[symbol] | /api/stock/[symbol]/history

↕ Node.js HTTP

EXTERNAL

Yahoo Finance API (via yahoo-finance2)

Real-time quotes (NSE/BSE) | Historical OHLCV data | Market summaries

Symbol search

↕

STORAGE

Browser localStorage (Client-side only)

Watchlist (saved symbols) | Theme preference

SECTION 04

Project Folder Structure

```
shares-app/
  ├── src/
  │   ├── app/              ← Next.js App Router root
  │   │   ├── page.tsx       ← Home Page (indices + movers)
  │   │   ├── layout.tsx     ← Root layout with Navbar + ThemeProvider
  │   │   ├── globals.css    ← CSS custom properties + base styles
  │   │   ├── api/           ← Serverless API Routes
  │   │   │   ├── live/       ← SSE streaming endpoint
  │   │   │   ├── market/     ← Nifty, Sensex, BankNifty
  │   │   │   ├── movers/     ← Top gainers, losers, trending
  │   │   │   ├── search/     ← Stock symbol search
  │   │   │   └── stock/[symbol]/ ← Quote + history for a symbol
  │   │   ├── stock/[symbol]/   ← Individual stock detail page
  │   │   ├── top-stocks/       ← Top gainers/losers/undervalued
  │   │   └── watchlist/        ← User's saved stocks
  │   ├── components/
  │   │   ├── Navbar.tsx     ← Navigation bar + search
  │   │   └── ThemeProvider.tsx ← next-themes wrapper
  │   └── lib/
  │       ├── scoring.ts     ← Stock scoring/rating algorithm
  │       ├── stocks-list.ts  ← Master list of 200+ Indian stocks
  │       ├── useLivePrice.ts ← Custom SWR hook for live prices
  │       └── utils.ts       ← cn(), formatCurrency(), isPositive()
  ├── next.config.js        ← Next.js configuration
  ├── tailwind.config.ts    ← Tailwind CSS config
  ├── tsconfig.json         ← TypeScript compiler options
  └── package.json          ← Dependencies
```

SECTION 05

API Routes Reference

METHOD	ENDPOINT	DESCRIPTION	DATA SOURCE
GET	/api/market	Returns live data for Nifty 50, Sensex, Bank Nifty, and Nifty IT indices	yahoo-finance2
GET	/api/movers	Returns today's top gainers, top losers, and trending Indian stocks	yahoo-finance2
SSE	/api/live	Server-Sent Events stream. Pushes updated index prices every ~10 seconds to all connected clients	yahoo-finance2
GET	/api/search?q={query}	Searches for Indian stock symbols by name or ticker. Returns matching results	yahoo-finance2
GET	/api/stock/[symbol]	Returns full quote data for a specific stock — price, change, market cap, P/E ratio, volume, 52-week range	yahoo-finance2
GET	/api/stock/[symbol]/history	Returns historical OHLCV (Open/High/Low/Close/Volume) data for charting. Accepts period query param (1wk, 1mo, 3mo, 1y)	yahoo-finance2

SECTION 06

Application Workflow

1

User Opens the App

The browser requests the Next.js home page. The root `layout.tsx` renders the **Navbar** (with search, theme toggle, and navigation links) and wraps content in the **ThemeProvider** which applies the stored dark/light preference.

Next.js SSR | ThemeProvider | Navbar.tsx

2

Initial Data Fetch

On mount, `page.tsx` fires two parallel REST calls: **/api/market** (Nifty, Sensex indices) and **/api/movers** (top gainers/losers/trending). The server-side API routes call **yahoo-finance2** to fetch live data from Yahoo Finance and return structured JSON.

fetch() | /api/market | /api/movers | yahoo-finance2

3

SSE Live Connection Established

Simultaneously, the client opens a persistent **Server-Sent Events** connection to **/api/live**. The server streams live index price updates every ~10 seconds as JSON events. The client listens for `indices` events and updates state, triggering flash animations (green/red) on changed prices.

EventSource API | /api/live (SSE) | Flash Animation

4

Movers Auto-Refresh

A `setInterval` runs every **30 seconds** to silently re-fetch movers data in the background. A spinner indicator shows when a refresh is in progress. The user can also manually refresh via the **RefreshCw** icon button.

setInterval | Background Polling | 30s interval

5

User Searches for a Stock

The user types in the Navbar search box. Debounced input fires a call to **/api/search?q=**, which uses *yahoo-finance2.search()* to find matching Indian stocks. Results appear as a dropdown. Clicking navigates to the Stock Detail page.

Debounce | /api/search | Navbar.tsx

6

Stock Detail Page

The dynamic route `/stock/[symbol]` renders. It calls **/api/stock/[symbol]** to get current fundamentals (price, P/E, EPS, market cap, 52W range). A **Recharts** Area chart displays historical price data from **/api/stock/[symbol]/history**. Users can toggle between 1W, 1M, 3M, and 1Y views. A **scoring algorithm** (lib/scoring.ts) rates the stock based on fundamentals.

Dynamic Route | Recharts | SWR | Scoring Algorithm

7

Watchlist Management

Users can click the **Star** ⭐ button on any stock to save it. The watchlist is stored in **browser localStorage** — no backend database needed. The `/watchlist` page reads these symbols, fetches live quotes for each, and displays them as a live dashboard.

localStorage | Watchlist Page | No Database

8

Theme Toggle

The theme toggle button in the Navbar switches between dark and light mode via **next-themes**. CSS custom properties (variables) defined in *globals.css* ensure all colors update seamlessly. The preference is persisted in localStorage for the next visit.

next-themes | CSS Variables | localStorage

SECTION 07

📦 Dependencies Summary

PACKAGE	VERSION	CATEGORY	PURPOSE
`next`	14.2.5	Framework	SSR, routing, API routes, image optimization
`react`	^18	UI	Component model, hooks, virtual DOM
`react-dom`	^18	UI	React renderer for the browser
`typescript`	^5	Language	Type safety, interfaces, enums
`tailwindcss`	^3.4.1	Styling	Utility-first CSS framework
`yahoo-finance2`	^3.13.2	Data	Real-time stock data from Yahoo Finance
`framer-motion`	^11.3.19	Animation	Declarative UI animations and transitions
`recharts`	^2.12.7	Chart	SVG-based interactive charts for price history
`swr`	^2.2.5	Data Fetching	Stale-while-revalidate caching for React
`lucide-react`	^0.427.0	Icons	Consistent SVG icon library
`next-themes`	^0.3.0	Theming	Dark/light mode management
`axios`	^1.7.3	HTTP	Promise-based HTTP client
`clsx`	^2.1.1	Utility	Conditional class name joining
`tailwind-merge`	^2.4.0	Utility	Merge Tailwind classes without conflicts

PACKAGE	VERSION	CATEGORY	PURPOSE
`postcss`	^8	Build	CSS transformation pipeline for Tailwind
`autoprefixer`	^10.0.1	Build	Automatic CSS vendor prefixes
`eslint`	^8	Dev	Code linting and quality enforcement

Shares App — Indian Stock Intelligence Platform

Generated: March 2026 · Built with Next.js 14, React 18, TypeScript & Tailwind CSS